4.2 Writing Equations in Point-Slope Form

Name _____ Period ____ Date _____

Algebra I 4.2 (30-minute plan) · Student sheet

I can…

- write an equation of a line given its slope and a point.
- write an equation of a line given two points.
- use point-slope form to write linear models.

Vocabulary · fill in the missing word

Point-slope form — $y - y_1 = m(x - x_1)$, for a line _____ (x_1, y_1) with slope m

Part 1 · Notes and examples (fill in each note, then work the example)

Point-slope form: Point-slope form: _____.

Reading it: (x_1, y_1) is a _____ on the line, and m is the _____.

Example: $y + 2 = 3(x - 4)$ passes through _____.

Example 1

Write an equation in point-slope form of the line through $(4, -3)$ with slope -2.

Example 2

A plumber charges a fixed fee plus an hourly rate. A 2-hour job costs $190 and a 5-hour job costs $385. Write an equation for the cost C of an h-hour job.

Part 2 · Partner practice

Partner practice

Write an equation in point-slope form through $(-5, 2)$ with slope $\frac{1}{2}$.